Adrosonic Phase 2 - Final Evaluation Report

Evaluation date: 2026-10-03 **Project:** Vector Database Design for Large-Scale Precision Retrieval in RAG Systems
Execution status: Blocked before corpus ingestion; only offline/unit-level checks were measurable.

1. Executive summary

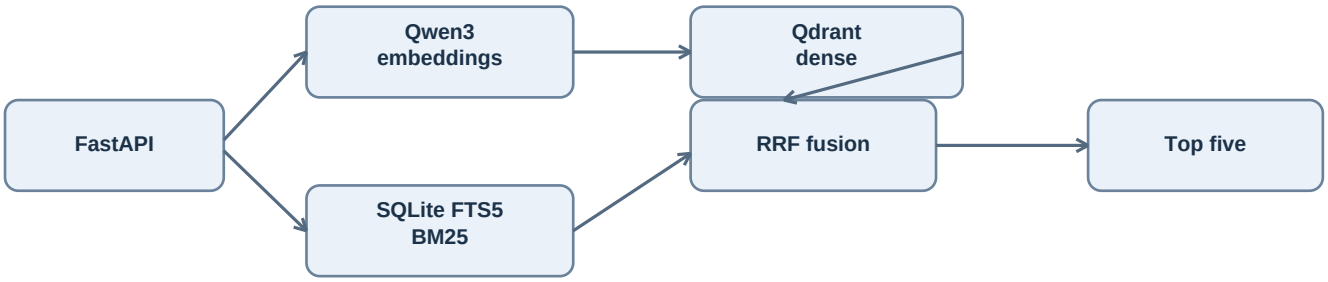
The ingestion command was actually run with `--limit 100000`, but Hugging Face access failed with a connection-refused error before the dataset yielded a row. The installed environment also lacks `qdrant-client` and RAGAS, and the Qwen checkpoint is not cached. The configured Qdrant and SQLite BM25 stores are absent. Therefore the verified corpus size is **0**; no real retrieval queries, Mistral judge calls, RAGAS scores, or retrieval latency samples were produced.

This is a blocked execution report, not a successful 100k evaluation. No missing metric is represented as zero or inferred. The project was minimally updated with an official MS MARCO Parquet loading route and an independent corpus validation command. Offline tests passed for the components available in this environment. Detailed cause and follow-up evidence are in ingestion_diagnostics.md.

2. Problem statement and Phase 2 objectives

Phase 2 is intended to index 100,000 original MS MARCO passages with Qwen3-Embedding-0.6B in persistent local Qdrant, maintain the same corpus in SQLite FTS5 BM25, fuse dense and sparse results with RRF, compare retrieval on valid MS MARCO judgments, evaluate Context Precision/Recall with RAGAS and an explicitly configured Mistral judge, and benchmark 100 dense plus 100 hybrid requests. This run could not progress past source acquisition.

3. System architecture



MS MARCO passages feed the embedding and lexical indexes; search fuses ranked results.

The architecture above describes the inspected implementation, not a live end-to-end run. This run did not instantiate Qdrant or Qwen.

4. Actual hardware and software environment

Component	Observed value
OS	Windows 11, build 10.0.26300
Python	3.13.15
CPU	20 logical CPUs
RAM	7.7 GB
Accelerator	CPU only; PyTorch reports no CUDA device
Free disk at audit	213.73 GB on C:
PyTorch / Transformers / Datasets	2.14.0 / 5.8.1 / 5.0.0
FastAPI / Pydantic	0.119.0 / 2.11.7
Streamlit / pytest / httpx / psutil	1.48.0 / 8.4.2 / 0.28.1 / 7.0.0
qdrant-client / RAGAS	Not installed
Qwen model cache	Absent

5. MS MARCO dataset source and corpus statistics

Configured dataset: `microsoft/ms_marco`, config `v1.1`; `train` is the indexing split and `validation` is the intended evaluation split. The loader uses the repository's v1.1 Parquet route to avoid the legacy dataset-script pathway under Datasets 5.x. Dataset version/commit was not pinned; the loader references `main`.

Corpus measure	Verified result
Requested limit	100,000
Source records yielded	0
Passage records read	0
Unique passage IDs indexed	0
Nonempty passages indexed	0
Duplicates / malformed passages observed	0 / 0 (iteration never began)
Qdrant logical points	Unavailable; store absent and client missing
BM25 logical documents	0; configured SQLite store absent
Dense/sparse IDs equal	Not evaluated

The source request failed with `httpx.ConnectError: [WinError 10061] No connection could be made because the target machine actively refused it`. No data was downloaded or substituted.

6. Actual indexing results

`python -m app.ingest --limit 100000` was executed. Ingestion failed while opening the dataset, before constructing the embedder or index. It reported no silent skips: zero rows were yielded and zero passages were processed. No duration, throughput, disk usage attributable to indexing, or index memory was measured.

Store	Configured location	Actual status/count
Qdrant	`storage/qdrant`, collection `passages`	Directory/collection absent; count API unavailable
SQLite FTS5 BM25	`storage/sparse.sqlite3`	File absent; 0 documents
Update journal	`storage/update_journal.jsonl`	Absent

7. Qwen3 embedding configuration

Setting	Configured value	Observed execution
Model	`Qwen/Qwen3-Embedding-0.6B`	Not loaded; cache absent and source acquisition failed first
Revision	`main`	Mutable branch, not a pinned commit
Device	`auto`	Host is CPU-only; model device selection was not reached
Precision	`auto`	No effective dtype measured; loader currently uses the model's default dtype
Batch size	4	Configured; no batch executed
Vector dimension	Model hidden size intended	Not measured
Normalization	L2 normalization in source	Not executed
Embedding errors	N/A	No embeddings attempted

The source implementation loads the model once per embedder, uses inference mode, separates query instruction formatting from document encoding, pools the final non-padding token, and normalizes vectors. Those behaviors were inspected in code, not validated against live model outputs.

8. Qdrant configuration

Configured local persistent collection is `passages` at `storage/qdrant`, with cosine distance, HNSW `m=16`, `ef_construct=100`, and search `ef_search=64`. The source includes payload indexes for selected metadata. `qdrant-client` could not be installed from the available package index, and the storage directory did not exist. Collection vector size, distance, payload indexes, persistence, and restart behavior were not verified.

9. BM25 configuration

The inspected sparse implementation is SQLite FTS5 with `unicode61 remove_diacritics 2`, original text, stable logical IDs, and SQLite `bm25()` ranking. The configured path is `storage/sparse.sqlite3`. A fixture test exercised FTS retrieval, metadata filtering, updates, and deletes. No production database or corpus was created.

10. Hybrid RRF methodology

The inspected hybrid path requests up to 50 candidates from each retriever and combines one-based ranks with Reciprocal Rank Fusion using `k=60`; the default output size is five. RRF ordering and tie-breaking passed a unit test. No dense or hybrid retrieval ran against an actual corpus.

11. Actual retrieval queries and examples

Retrieval mode	Valid queries executed	Retrieved passages	Example
Dense	0	0	None available
BM25	0	0	None available
Hybrid RRF	0	0	None available

The validation split could not be downloaded, so no valid query, answer, or relevance judgments were available. No fabricated query or relevance label was used. The evaluation and benchmark directories contain status artifacts, not per-query result records.

12. Mistral RAGAS evaluation setup

The configured judge model is `mistral-small-latest`; the project has a Mistral integration and a Mistral/RAGAS evaluation path. The environment reports a Mistral key configured, but the key was never displayed or written to artifacts. No Mistral API request was made because free-tier/model allowance was not independently verified (`MISTRAL_FREE_TIER_CONFIRMED=false`). RAGAS is not installed. There were **0** Mistral calls and no RAGAS LLM factory invocation in this run.

13. Actual Context Precision and Context Recall

Retrieval mode	Evaluation queries	Context Precision	Context Recall	Status
Dense baseline	0	N/A	N/A	Not evaluated
Hybrid RRF	0	N/A	N/A	Not evaluated

Targets are Context Precision > 0.75 and Context Recall > 0.70. Neither was tested. Unavailable does not mean zero. No per-query RAGAS rows or evaluation exceptions from metric execution exist.

14. Dense versus hybrid comparison

Measure	Dense	Hybrid RRF
Same valid query set	Not available	Not available
Retrieved candidates/results	0	0
Context Precision / Recall	N/A / N/A	N/A / N/A
Comparative conclusion	No evidence for a quality comparison	No evidence for a quality comparison

Recall@5, MRR@5, and nDCG@5 were not computed; the valid qrels/query set was unavailable.

15. Supplementary retrieval metrics

No ranking metrics can be computed without actual retrieved IDs and relevance judgments. No values are estimated or fabricated.

16. Actual 100-query benchmark results

The request was for 100 consecutive requests in each of dense and hybrid modes. None ran because no corpus or valid query set was available.

Mode	Requests	Failed	p50 ms	p95 ms	p99 ms	Mean ms	Min ms	Max ms	p95 < 300 ms
Dense	0	0	N/A	N/A	N/A	N/A	N/A	N/A	Unverified
Hybrid	0	0	N/A	N/A	N/A	N/A	N/A	N/A	Unverified

17. Latency breakdown and target status

No latency samples exist. Query embedding, Qdrant, BM25, RRF, serialization, and end-to-end API latencies are all unavailable; Mistral generation latency was not measured. The inspected benchmark currently measures total in-process search time and does not expose all requested stage timings or API network overhead. The p95 target of 300 ms was not tested and is not claimed met.

18. Index consistency and persistence validation

No indexes existed, so corpus ID equality, exact Qdrant count, sparse count, nonempty stored texts, sample retrieval, live search, and reopen-after-restart could not be measured. Added `python -m app.validate_corpus` to compare complete Qdrant logical ID payloads against the SQLite ID set, verify expected counts/texts and sample passage retrieval, and attempt live model/vector/search checks. It writes `corpus_validation.json` and fails closed when prerequisites or acceptance checks fail. The current recorded status is blocked.

19. Automated test results

Command/check	Actual result
`python -m compileall -q app streamlit_app.py tests`	Passed
`python -m pytest tests -p no:cacheprovider -q`	8 passed, 2 skipped; exit code 0
Skips	Qdrant integration/update/filter/reopen tests; `qdrant-client` unavailable
API degraded-index behavior	Search/upsert/delete return 503 after exception handling fix
SQLite FTS5 fixture and RRF	Passed
Live API results from indexed corpus	Not available; no corpus

The test evidence is in test_results.md. After adding the validation script, it is syntax-checked as part of the final compile command; no new tests were added.

20. Errors encountered and fixes attempted

1. Datasets 5.x cannot use the legacy dataset-script pathway for this source. The loader now requests the official v1.1 repository Parquet file directly.
2. The Parquet request failed with `WinError 10061` because the host actively refused the Hugging Face connection. This environmental blocker was not bypassed and no substitute data was used.
3. Installing `qdrant-client` failed with ?No matching distribution found?; no collection was initialized.
4. The Qwen checkpoint was not locally cached; model download/inference was not reached.
5. Missing-index API mutations previously converted HTTP 503 readiness errors into 500 responses. The handlers now re-raise `HTTPException`; the degraded-mode test passes.

21. Limitations and unfulfilled requirements

The mandatory 100,000-passage corpus was not indexed. Qdrant/BM25 parity, retrieval, 20-query RAGAS evaluation, actual Mistral judge usage, the 100-query-per-mode benchmark, p95 latency, supplementary metrics, persistence reopening, and real API retrieval remain incomplete. The active runtime uses Python 3.13.15 and has limited RAM (7.7 GB) with no GPU; the project recommends Python 3.11/3.12 for dependency compatibility. The model revision is not commit-pinned. No claim is made that a configured feature passed a live integration check.

22. Reproduction instructions

On a machine where the official Hugging Face dataset/model endpoints and package index are reachable, use Python 3.11 or 3.12 and install the pinned project requirements in a fresh environment. Preserve the existing secret file; never place key values in command output.

```
py -3.12 -m venv .venv
.\.venv\Scripts\Activate.ps1
python -m pip install --upgrade pip
pip install -r requirements.txt
python -m app.ingest --limit 100000
python -m app.validate_corpus
python -m app.prepare_eval --count 20
python -m app.evaluate
```

```
python -m app.benchmark --queries 100
python -m pytest tests -p no:cacheprovider -q
```

Proceed to evaluation only after `corpus_validation.json` reports `passed`. Verify that the configured Mistral model and account allowance permit use before setting `MISTRAL_FREE_TIER_CONFIRMED=true`; keep the key out of logs and reports. Launch the services with `uvicorn app.api:app --host 127.0.0.1 --port 8000` and `streamlit run streamlit_app.py` after successful indexing.

Acceptance checklist

Requirement	Implemented	Tested here	Measured from real corpus	Status
MS MARCO v1.1 Parquet loader	Yes	Loader configuration test	No rows yielded	Blocked by network
Exactly 100,000 unique passages indexed	Ingestion path exists	No	0 verified	Not met
Qwen document/query vectors	Source implementation exists	No live model test	No vectors	Not met
Persistent Qdrant collection	Source implementation exists	Qdrant integration skipped	No collection	Not met
SQLite FTS5 BM25	Yes	Fixture tests passed	No corpus database	Partial
Dense/sparse logical ID validator	Added	Syntax check only; no stores	Cannot run to completion	Blocked
20+ valid dense/hybrid retrieval queries	Pipeline exists	No	0	Not met
RAGAS Context Precision/Recall with Mistral	Integration path exists	No; RAGAS absent and allowance unverified	No scores, 0 calls	Not met
100 dense + 100 hybrid latency queries	Benchmark script exists	No	0	Not met
p95 below 300 ms	Target documented	No	N/A	Unverified
Automated offline tests	Yes	8 passed, 2 skipped	As shown above	Partial
PDF and reproducibility artifacts	Yes	PDF text/render inspection	Reflect blocked run	Complete for reporting